

Assignment 1 - Theory of Dynamic System.

Exercise 1

$$\rightarrow \therefore A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

A is an upper triangular Matrix, $\therefore$ the eigenvalues are on its diagonal.

$$\therefore \boxed{\lambda_1 = 3, \lambda_2 = -4, \lambda_3 = -1} \quad \text{--- (1)}$$

Now, for the eigenvectors, we use the equation $(A - \lambda I)\vec{v} = 0$ for each eigenvalue:

from (1), $\lambda_1 = 3$

$$\therefore [A - [3]I]\vec{v}_1 = \left[\begin{bmatrix} 3 & -1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & -1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right] \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 0 & -1 & 0 \\ 0 & -7 & 2 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore 0 - y_1 = 0, y_1 = 0 \quad \text{--- (2)}$$

$$\therefore 0 - 7y_1 + 2z_1 = 0, \text{ from (2), } z_1 = 0 \quad \text{--- (3)}$$

$$\therefore -4z_1 = 0, z_1 = 0 \quad \text{--- (4)}$$

$\therefore x_1$ can be any non-zero scalar, let $x_1 = 1$ — (5)

$\therefore$ from (2)(3)(5), $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ — (6)

Now, from (1), $\lambda_2 = -4$,

$$[A - [-4]I] \vec{v}_2 = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} + 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \vec{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore 7x_2 - y_2 = 0 \text{ — (7)}$$

$$\therefore 2z_2 = 0, z_2 = 0 \text{ — (8)}$$

$\therefore$ Let $x_2 = 1$, from (7), $y_2 = 7$ — (10)
— (9)

$\therefore$ from (8)(9)(10), $\vec{v}_2 = \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix}$

Now, from (1) $\lambda_3 = -1$

$$[A - [-1]I] \vec{v}_3 = 0$$

$$\therefore \left[\begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} - [-1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right] \vec{V}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 4 & -1 & 0 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore 4x_3 - y_3 = 0 \quad \text{--- (11)}$$

$$\therefore -3y_3 + 2z_3 = 0 \quad \text{--- (12)}$$

$$\text{from (11) (12), } z_3 = 6x_3 \quad \text{--- (13)}$$

$$\text{let } x_3 = 1 \quad \text{--- (14), from (11) (14), } y_3 = 4 \quad \text{--- (15)}$$

$$\text{from (14) (13), } z_3 = 6 \quad \text{--- (16)}$$

$$\therefore \text{from (14) (15) (16), } \vec{V}_3 = \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix}$$

Now for the stability analysis:

$\therefore$ Stability of a linear dynamical system is determined by the real parts of the eigenvalues of Matrix A.

We have, $\lambda_1 = 3, \lambda_2 = -4, \lambda_3 = -1$

$\therefore$ There is at least one eigenvalue ($\lambda_1 = 3$) with a positive real part, the dynamical system is Unstable

Exercise 2:

$$\rightarrow \therefore A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Now for the eigenvalues, the characteristic equation for finding eigenvalues of a square matrix A is: $\det[A - \lambda I] = 0$

$$\therefore A - \lambda I = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 0 & 1 \\ -1 & 1-\lambda & -1 \\ 0 & 1 & -\lambda \end{bmatrix}$$

$$\therefore \det[A - \lambda I] = +[1-\lambda][1-\lambda](-\lambda) - [-1][1](-\lambda) - 0 + [1][1](-\lambda) - [-1-\lambda](-\lambda)$$

$$= -\lambda^3 + 2\lambda^2 - 2\lambda = 0$$

$$\lambda[\lambda^2 - 2\lambda + 2] = 0, \therefore \text{One eigen value } \boxed{\lambda_1 = 0} \text{ --- (1)}$$

Now, $\lambda^2 - 2\lambda + 2$, using, $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\therefore \lambda = \frac{-[-2] \pm \sqrt{[-2]^2 - 4[1][2]}}{2[1]}$$

$$\therefore \lambda = 1 \pm i, \therefore \boxed{\lambda_2 = 1+i} \text{ --- (2)}, \boxed{\lambda_3 = 1-i} \text{ --- (3)}$$

Now for the eigenvectors, we use $[A - \lambda I]\vec{v} = 0$ for each eigenvalues.

For $\lambda_1 = 0$,

$$\therefore [A - [0]I] \vec{v}_1 = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore x_1 + z_1 = 0 \quad \text{--- (4)}$$

$$\therefore -x_1 + y_1 - z_1 = 0 \quad \text{--- (5)}$$

$$\therefore y_1 = 0 \quad \text{--- (6)}$$

$$\text{Let } x_1 = 1 \quad \text{--- (7)}, \text{ from (4) (7), } z_1 = -1 \quad \text{--- (8)}$$

$$\text{from (5) (7) (8), } \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Now for $\lambda_2 = 1+i$,

$$\therefore [A - [1+i]I] \vec{v}_2 = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 1+i & 0 & 0 \\ 0 & 1+i & 0 \\ 0 & 0 & 1+i \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 & 1 \\ -1 & i & -1 \\ 0 & 1 & -1-i \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore -ix_2 + z_2 = 0 \quad \text{--- (9)}$$

$$\therefore -x_2 + iy_2 - z_2 = 0 \quad \text{--- (10)}$$

$$\therefore y_2 - 1 - z_2 = 0 \quad \text{--- (11)}, \text{ Let } x_2 = 1 \quad \text{--- (11)}, \text{ from (9) (11), } z_2 = i \quad \text{--- (12)}$$

$$\text{from (11) (12) (10), } y_2 = -1+i \quad \text{--- (13)}$$

from (11)(12)(13), $\vec{v}_2 = \begin{bmatrix} 1 \\ -1+i \\ i \end{bmatrix}$ — (14)

Now for $\lambda_3 = 1 - i$, the eigenvector $\vec{v}_3$ is the complex conjugate of $\vec{v}_2$, $\therefore$ from (14), $\vec{v}_3 = \overline{\vec{v}_2} = \begin{bmatrix} 1 \\ -1-i \\ -i \end{bmatrix}$

$\therefore \vec{v}_3 = \begin{bmatrix} 1 \\ -1-i \\ -i \end{bmatrix}$

Now for the Stability Analysis;

$\therefore$ Stability of the ~~dynam~~ dynamical system is determined by the real parts of its eigenvalues

$\therefore \text{Re}[\lambda_1] = \text{Re}[0] = 0$

$\text{Re}[\lambda_2] = \text{Re}[1+i] = 1$

$\text{Re}[\lambda_3] = \text{Re}[1-i] = 1$

$\therefore$ There exists at least one eigenvalue with a positive real part ($\text{Re}[\lambda_2]$ and $\text{Re}[\lambda_3] > 0$),

$\therefore$ The system is Unstable

Exercise 3

$$\rightarrow \therefore A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 0 & 1 \end{bmatrix}$$

$\therefore A$ is an upper triangular Matrix, its eigenvalues are its diagonals,

$$\therefore \lambda_1 = 3, \lambda_2 = -4, \lambda_3 = -1 \quad \text{--- (1)}$$

Now for the Controllability Analysis:

$\therefore$ An eigenvalue λ_i is controllable if $\text{rank} \begin{bmatrix} A - \lambda_i I & B \end{bmatrix} = n$

$\therefore n = 3$ here

$$\therefore \begin{bmatrix} A - 3I & B \end{bmatrix} = \left[\begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right] \left| \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right|$$

$$= \begin{bmatrix} 0 & -1 & 0 & 1 \\ 0 & -7 & 2 & 1 \\ 0 & 0 & -4 & 0 \end{bmatrix}$$

$\therefore$ Converting to reduced row echelon form,

$$\therefore -R_1 \rightarrow -R_1$$

$$R_2 \rightarrow R_2 + 7R_1$$

$$R_2 \rightarrow \frac{1}{2}R_2$$

$$R_3 \rightarrow R_3 + 4R_2$$

$$R_3 \rightarrow -\frac{1}{2}R_3$$

$$R_1 \rightarrow R_1 + R_3$$

$$R_2 \rightarrow R_2 + 3R_3$$

$$\rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{Rank} = 3 \quad \text{--- (2)}$$

from ③, last three columns are linearly independent,

$$\therefore \lambda_1 = 3 \text{ is controllable} \text{ --- ③}$$

Now for $\lambda_2 = -4$

$$\begin{aligned} \therefore [A + 4I | B] &= \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} + 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left| \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right. \\ &= \begin{bmatrix} 0 & -1 & 0 & 1 \\ 0 & -7 & 2 & 1 \\ 0 & 0 & -4 & 0 \end{bmatrix} \end{aligned}$$

Converting to RREF:

$$R_1 \rightarrow 1/7 R_1$$

$$R_2 \rightarrow 1/2 R_2$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$R_3 \rightarrow -\frac{2}{3} R_3$$

$$R_1 \rightarrow R_1 - 1/7 R_3$$

$$R_2 \rightarrow R_2 - 1/2 R_3$$

$$= \begin{bmatrix} 1 & -1/7 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \therefore \text{Rank} = 3 \text{ --- ④}$$

$\therefore$ from ④, All three rows are linearly independent,

$$\therefore \lambda_2 = -4 \text{ is controllable} \text{ --- ⑤}$$

Now for $\lambda_3 = -1$:

$$[A+I \mid B] = \begin{bmatrix} 4 & -1 & 0 & 1 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Converting it to RREF:

$$R_1 \rightarrow 1/4 R_1$$

$$R_2 \rightarrow -1/3 R_2$$

$$R_1 \rightarrow R_1 + 1/4 R_2$$

$$= \begin{bmatrix} 1 & 0 & -1/6 & 1/8 \\ 0 & 1 & -2/3 & -1/3 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \therefore \text{Rank} = 2 \text{ --- (6)}$$

$\therefore$ from (6), above two rows are linearly independent,

$\therefore \lambda_3 = -1$ is not controllable --- (7)

Now for Observability Analysis, $\therefore C = [2 \ 0 \ 1]$

$\therefore$ An eigenvalue (λ_i) is observable if $\text{rank} \begin{bmatrix} A - \lambda_i I \\ C \end{bmatrix} = n$

$$\therefore \text{for } \lambda_1 = 3, \begin{bmatrix} A - 3I \\ C \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 7 & 2 \\ 0 & 0 & -4 \\ 2 & 0 & 1 \end{bmatrix}$$

Converting it to RREF:

$$R_1 \leftrightarrow R_4$$

$$R_2 \leftrightarrow R_4$$

$$R_1 \rightarrow \frac{1}{2}R_1$$

$$R_2 \rightarrow -R_2$$

$$R_3 \rightarrow -\frac{1}{4}R_3$$

$$R_4 \rightarrow R_4 + 7R_2$$

$$R_1 \rightarrow R_1 - \frac{1}{2}R_3$$

$$R_4 \rightarrow R_4 - 2R_3$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \therefore \text{Rank} = 3 \quad \text{--- (8)}$$

$\therefore$ from (8), above three rows are linearly independent,

$\therefore \lambda_1 = 3$ is observable --- (9)

Now for $\lambda_3 = -\frac{1}{2}$, $\therefore [A + \frac{1}{2}I] = \begin{bmatrix} 4 & -1 & 0 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

Converting it to RREF:

$$R_4 \rightarrow R_4 - \frac{1}{2}R_1$$

$$R_3 \leftrightarrow R_4$$

$$R_3 \rightarrow R_3 + \frac{1}{6}R_2$$

$$R_1 \rightarrow \frac{1}{4}R_1$$

$$R_2 \rightarrow -\frac{1}{3}R_2$$

$$R_3 \rightarrow \frac{3}{4}R_3$$

$$R_2 \rightarrow R_2 + \frac{2}{3}R_3$$

$$R_1 \rightarrow R_1 + \frac{1}{4}R_2$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \therefore \text{Rank} = 3 \quad \text{--- (10)}$$

$\therefore$ from (10), above three rows are linearly independent,

$\therefore \lambda_3 = -1$ is observable --- (11)

Now for $\lambda_2 = -4$

$$\left[\frac{A+4I}{C} \right] = \begin{bmatrix} 7 & -1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \\ 2 & 0 & 1 \end{bmatrix}$$

Converting it to RREF:

$$\therefore R_4 \rightarrow R_4 - 2/7 R_1$$

$$R_2 \leftrightarrow R_4$$

$$R_4 \rightarrow R_4 - 2/3 R_3$$

$$R_1 \rightarrow 1/7 R_1$$

$$R_2 \rightarrow 7/2 R_2$$

$$R_3 \rightarrow 1/3 R_3$$

$$R_2 \rightarrow R_2 - 7/2 R_3$$

$$R_1 \rightarrow R_1 + 1/7 R_2$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \therefore \text{Rank} = 3$$

$\therefore$ from (12), above three rows are linearly independent

$\therefore \lambda_2 = -4$ is observable — (13)

Now for transfer function calculation,

$$\therefore \text{The transfer function : } W[s] = C[sI - A]^{-1} B$$

Let find $[sI - A]^{-1}$:

$$= \left[\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} \right]^{-1} = \begin{bmatrix} s-3 & 1 & 0 \\ 0 & s+4 & -2 \\ 0 & 0 & s+1 \end{bmatrix}^{-1}$$

Now, for $\text{adj}[sI - A] = \text{transpose of cofactor Matrix}$

$$\therefore \text{adj}[sI - A] = C^T$$

$$\therefore C_{11} = \det \begin{vmatrix} s+4 & -2 \\ 0 & s+1 \end{vmatrix} = (s+4)(s+1)$$

$$C_{12} = -\det \begin{vmatrix} 0 & -2 \\ 0 & s+1 \end{vmatrix} = 0$$

$$C_{13} = \det \begin{vmatrix} 1 & 0 \\ 0 & s+1 \end{vmatrix} = 0$$

$$C_{21} = -\det \begin{vmatrix} 1 & 0 \\ 0 & s+1 \end{vmatrix} = -[s+1]$$

$$C_{22} = \det \begin{vmatrix} s-3 & 0 \\ 0 & s+1 \end{vmatrix} = [s-3][s+1]$$

$$C_{23} = -\det \begin{vmatrix} s-3 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{31} = \det \begin{vmatrix} 1 & 0 \\ s+4 & -2 \end{vmatrix} = -2$$

$$C_{32} = -\det \begin{vmatrix} s-3 & 0 \\ 0 & -2 \end{vmatrix} = 2[s-3]$$

$$C_{33} = \det \begin{vmatrix} s-3 & 1 \\ 0 & s+4 \end{vmatrix} = [s-3][s+4]$$

$$\therefore C = \begin{bmatrix} [s+4][s+1] & 0 & 0 \\ -[s+1] & [s-3][s+1] & 0 \\ -2 & 2[s-3] & [s-3][s+4] \end{bmatrix}$$

$$\therefore \text{adj}[sI-A] = C^T = \begin{bmatrix} [s+4][s+1] & -[s+1] & -2 \\ 0 & [s-3][s+1] & 2[s-3] \\ 0 & 0 & [s-3][s+4] \end{bmatrix} \quad (14)$$

$$\text{Now, } \det[sI-A] = [s-3][s+4][s+1] \quad (15)$$

$$\therefore [sI-A]^{-1} = \frac{1}{\det[sI-A]} \text{adj}[sI-A], \text{ from (14) (15)}$$

$$[sI-A]^{-1} = \frac{1}{[s-3][s+4][s+1]} \begin{bmatrix} [s+4][s+1] & -[s+1] & -2 \\ 0 & [s-3][s+1] & 2[s-3] \\ 0 & 0 & [s-3][s+4] \end{bmatrix} \quad (16)$$

$$\text{Now for } W[s] = C[sI-A]^{-1}B :$$

$$W[s] = \frac{1}{[s-3][s+4][s+1]} \begin{bmatrix} 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} [s+4][s+1] & -[s+1] & -2 \\ 0 & [s-3][s+1] & 2[s-3] \\ 0 & 0 & [s-3][s+4] \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{[s-3][s+4][s+1]} \begin{bmatrix} 2[s+4][s+1] \cdot -2[s+1] \cdot -4 + [s-3][s+4] \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$W[s] = \frac{2[s+3]}{[s-3][s+4]}, \therefore \text{The poles of the system are at } \boxed{s=3} \text{ and } \boxed{s=-4}$$

(17)

Now for Motivation of Shape:

from (3)(9), $\lambda_1 = 3$: Controllable and Observable $\rightarrow$ Pole (18)

from (4)(13), $\lambda_2 = -4$: Controllable and Observable $\rightarrow$ Pole (19)

from (5)(11), $\lambda_3 = -1$: Not Controllable and not Observable $\rightarrow$ No pole (20)

$\therefore$ (18)(19) matches the poles, calculated in (17) transfer function, and it is uncontrollable at $s = -1$

Now Analysis with $C = [0 \ 0 \ 1]$:

Here we only need to evaluate observability and transfer function

$$\therefore \text{for } \lambda_1 = 3: \left[\frac{A - 3I}{C} \right] = \begin{bmatrix} 0 & -1 & 0 \\ 0 & -7 & 2 \\ 0 & 0 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ The first column is all zeros, therefore rank is at most 2

$\therefore$ At $\lambda_1 = 3$, it is not observable (21)

$$\text{for } \lambda_2 = -4, \left[\frac{A + 4I}{C} \right] = \begin{bmatrix} 7 & -1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Converting to RREF

$$\begin{aligned} R_1 &\rightarrow \frac{1}{7} R_1 \\ R_2 &\leftrightarrow R_1 \\ R_3 &\rightarrow R_3 - 3R_2 \\ R_4 &\rightarrow R_4 - 2R_2 \end{aligned} \quad = \quad \begin{bmatrix} 1 & -1/7 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \text{rank} = 2$$

$\therefore$ We have two rows with all zeros

$\therefore$ At $\lambda_2 = -4$, it is not observable — (22)

for $\lambda_3 = -1$: $\left[\frac{A+I}{C} \right] = \begin{bmatrix} 4 & -1 & 0 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Converting it to RREF:

$$\begin{aligned} R_3 &\leftrightarrow R_1 \\ R_1 &\rightarrow \frac{1}{4} R_1 \\ R_2 &\rightarrow -\frac{1}{3} R_2 \end{aligned} \quad = \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \therefore \text{Rank} = 3$$

$R_2 \rightarrow R_2 + \frac{2}{3} R_3$ At $\lambda_3 = -1$, it is observable — (23)

$R_1 \rightarrow R_1 + \frac{1}{4} R_2$

Now for the transfer function for $C = [0 \ 0 \ 1]$

$\therefore W[s] = C [sI - A]^{-1} B$, from (16)

$$= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \cdot \frac{1}{[s-3][s+4][s+1]} \begin{bmatrix} [s+4][s+1] - [s+1] \\ [s-3][s+1] \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$W[s] = 0$ — (24)

Now for the Motivation of Shape for $C = [0 \ 0 \ 1]$;

$\therefore$ At $\lambda_1 = 3$, from (3)(21), Controllable but Not Observable
 $\rightarrow$ No Pole. — (25)

$\therefore$ At $\lambda_2 = -4$, from (4)(22), Controllable but Not Observable,
 $\rightarrow$ No pole — (26)

$\therefore$ At $\lambda_3 = -1$, from (5)(23), Not Controllable but Observable,
 $\rightarrow$ No Pole — (27)

Since No eigenvalue is simultaneously Controllable and Observable, there are no poles, The system's input ~~that~~ has no effect on its output. Also proved by

$$W[s] = 0 \quad \text{— (28)}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Now for the transfer function for $C = [0 \ 0 \ 1]$

$$M[s] = C[sI - A]^{-1}B \quad \text{from (8)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \frac{1}{(s-3)(s+4)(s+1)} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{(s-3)(s+4)(s+1)}$$

$$M[s] = 0 \quad \text{— (29)}$$